Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Solutions Engineer, Forward Deployed Engineer and Data Scientist with 3 years of industry experience building and deploying production GenAI applications end to end (RAG, Azure OpenAI, LlamaIndex, pgvector, FastAPI, ReactJS, AWS) in a regulated financial services setting. Hands-on across machine learning, data engineering (Python, SQL, PostgreSQL), analytics (Pandas, Power BI) and MLOps (Docker, Jenkins/GitLab CI/CD), and a de facto team lead for up to 4 interns. Recently completed an MSc in Data Science at the University of Bristol (awaiting result), including a dissertation project with AstraZeneca, and placed in three UK hackathons and datathons. Seeking AI, machine learning, data science and analytics, data engineering, full stack or DevOps/MLOps roles in the UK.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Generative AI / Retrieval Augmented Generation (RAG) Engineer

- Designed and built **GenAI applications** end to end (ReactJS frontend, FastAPI backend, **RAG** logic, vector database and embeddings pipeline) using **LlamaIndex**, **Azure OpenAI GPT-3.5** and **Llama-3** to answer department-specific queries for Compliance, Client Service and Investment teams.
- Maintained department-wise **vector databases** (**pgvector**, Azure OpenAI embeddings) for **semantic search**; allocated models by cost, with Llama-3 self-hosted on a GPU **EC2** instance for Client Service, and guardrailed behaviour through system prompts versioned in Git.
- Built **AI agents** that call **Python functions** as tools (**tool calling**) to run SQL-backed fund-comparison calculations and plot them into PDF reports on user request.
- Developed a **Python** automation job (**Azure OpenAI**, earlier **AWS Textract**) on **AWS SageMaker** to parse **300+ daily emails** from credit rating agencies into a daily analyst report, **saving 60 manual analyst hours a month**.

Full Stack and Forward Deployed Engineer

- Worked directly with Client Service, Compliance and Investment stakeholders, building to their requirements; the only engineer alongside the manager in year 1, later a **de facto team lead** guiding 2–4 interns.
- Served the **ReactJS** frontend with **Nginx** on one AWS EC2 instance and the **FastAPI** microservices (**Docker**, uvicorn) on another, with **RBAC** through **LDAP** authentication.
- Recreated the **Power BI** dashboards in ReactJS, making the data available across the organization at lower cost than scaling Power BI licences.

Data Engineer

- Designed the entire **Normalized SQL Schema** (**PostgreSQL**) and developed **Optimized SQL** queries, **SQL Functions** and **Procedures** for preprocessing and efficient data insertion.
- Wrote **Python** ingestion scripts for the enterprise-licensed **Morningstar API**, refreshed fortnightly and monthly for every fund in the Indian mutual fund market (2023–2025), and integrated it with legacy data held in analysts' Excel files.

Data Analyst

- Analyzed customer data and drew insights using **Python Pandas**.
- Developed **Power BI** dashboards for fortnightly credit report analysis, migrating financial analysts from **Excel to Power BI** and improving analyst efficiency by **10 hours a week**.
- Identified customer buying patterns using **Market Basket Analysis** (**Apriori Algorithm**) for the marketing team.

MLOps / DevOps / Cloud Infrastructure

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** utilizing **EC2, S3, ECR** and **Lambda**, provisioning EC2 instances with security-team **CloudFormation** templates.
- Restricted app access to company private IPs with **AWS Security Groups** and deployed an **Nginx** reverse proxy, one of the fixes that took the app from failing to passing the security team's VAPT.
- Built the deployment automation with **Jenkins** and **GitLab** (**CI/CD pipeline**): a one-click deploy of the dev or prod branch for separate frontend and backend repositories, each in its own **Docker** container.

PROJECTS

- **University of Bristol MSc Dissertation with AstraZeneca:** CellLineSelector (team project; ranks best-suited cancer cell lines for target inclusion and exclusion genes, validated with AstraZeneca scientists. My focus: data preprocessing, scoring algorithm, RAG layer.) | GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app (108 LLM personas across 12 archetypes, vision-model photo scoring, Pydantic-validated LLM output. ReactJS, FastAPI.) | GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (team project; CNN/ViT/ResNet encoders vs LSTM/attention-based decoders for image captioning, benchmarked against BLIP/SigLIP and commercial VLMs.) | GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator (car trained using a CNN model).

ACHIEVEMENTS

- **HDFC AMC Best Team Project 2024 and 2025 (internal recognition):** rating-change email automation (2024) and department-specific RAG chatbot (2025).
- **Project Hack 27:** Rolls Royce Derby 4th Place winner. (Human centric data analytics tool predicting the winner from wellbeing survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – spec-driven AI dating app.) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyds Bank:** 3rd Place winner. (Kaggle credit card fraud detection problem.)

SKILLS

- **AI / ML:** Generative AI, RAG, LLM applications, tool calling, prompt engineering, Azure OpenAI, LlamaIndex, Llama-3, embeddings, pgvector, scikit-learn, CNN.
- **Languages:** Python (pandas, FastAPI, scikit-learn, Pydantic), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell.
- **Cloud / DevOps:** AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker, Textract, Security Groups), Docker, Jenkins, GitLab CI/CD, Nginx, Linux, LDAP (RBAC).
- **Databases:** PostgreSQL (schema design, functions and procedures, pgvector), SQLite.
- **Tools:** Jupyter, Anaconda, Power BI, Git, VS Code, Putty, Claude Code.